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Status	Finished
Started	Thursday, 6 November 2025, 12:45 PM
Completed	Thursday, 6 November 2025, 1:14 PM
Duration	29 mins 34 secs
Marks	2.50/3.00
Grade	2.50 out of 3.00 (83.34%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.50 out of 1.00

Let A and X be the sets

$$A = \{a, b, c\}$$

$$X = \{!, *, +, -\}.$$

In this problem we consider words written in the alphabet which is the union of the two sets: $A \cup X$. This means that a word in this alphabet can contain letters from both A and X .

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set A .
2. A word must end with a letter from the set A .
3. A word can not contain two adjacent letters from the set X .

For example the word **a+baa!c** is valid, but the words **a+baa+!c**, **!aaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let a_n equal the number of valid words of length n .

a) Find a recurrence relation satisfied by a_n :

$$3 \cdot a[n-1] + 12 \cdot a[n-2] = a[n]$$

Your last answer was interpreted as follows:

$$3 \cdot a_{n-1} + 12 \cdot a_{n-2} = a_n$$

The variables found in your answer were: $[a_{n-2}, a_{n-1}, a_n]$

Syntax help: To express the symbol " a_n " you should enter **a[n]**. For example, if you think the correct answer is $a_{n+1} - 2a_{n-3} = 1$, you should answer **a[n+1] - 2*a[n-3] = 1**.

b) Compute the value

$$a_1 = 3$$

Your last answer was interpreted as follows:

$$3$$

c) Compute the value

$$a_3 = 36$$

Your last answer was interpreted as follows:

$$36$$

d) Find the general solution to the recurrence relation satisfied by a_n .

$$a_n = \boxed{a[n] = A*((3-\sqrt{57})/(2)) + B*((3+\sqrt{57})/(2))}$$

Your last answer was interpreted as follows:

$$a_n = A \cdot \left(\frac{3 - \sqrt{57}}{2} \right) + B \cdot \left(\frac{3 + \sqrt{57}}{2} \right)$$

The variables found in your answer were: $[A, B, a_n]$

Syntax help: You are free to use constants in your answer. For example, if you think the correct answer is $2 \cdot n - A \cdot 3^n + B$, you should answer **2*n - A*3^n + B**.

Question 2

Correct

Mark 1.00 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle 1, 2, 3 in a counter-clockwise manner. The symmetry $r1$ is rotation by 120 degrees, sending vertex 1 to vertex 2. The symmetry $r2$ is rotation by 240 degrees. The element $t1$ is the reflection symmetry that fixes vertex 1, and likewise for $t2$ and $t3$.

The multiplication table for this group looks like

id	r1	r2	t1	t2	t3
r1	r2	id	t2	t3	t1
r2	id	r1	t3	t1	t2
t1	t3	.	.	.	r1
t2	t1	.	.	.	r2
t3	t2	.	.	.	id

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3 .

Syntax help: Remember that we compose symmetries from left to right. So if a and b are two symmetries, then the composite $a \cdot b$ is the symmetry that first applies a , and then b .

$t3 \cdot t1 =$	r2
$t1 \cdot t1 =$	id
$t2 \cdot t1 =$	r1
$t3 \cdot r2 =$	t1
$t1 \cdot r2 =$	t2
$t2 \cdot r2 =$	t3
$t3 \cdot t2 =$	r1
$t1 \cdot t2 =$	r2
$t2 \cdot t2 =$	id

Your last answer was interpreted as follows:

r2

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

r1

Your last answer was interpreted as follows:

t1

Your last answer was interpreted as follows:

t2

Your last answer was interpreted as follows:

t3

Your last answer was interpreted as follows:

r_1

Your last answer was interpreted as follows:

r_2

Your last answer was interpreted as follows:

id

Question 3

Correct

Mark 1.00 out of 1.00

Let $(P_7, \bullet_7)$ be the group of natural numbers coprime to 7 with group operation given by multiplication modulo 7. The multiplication table for this group looks like

1	2	3	4	5	6
2	4	6	1	3	5
3	.	.	5	.	4
4	1	5	2	6	3
5	.	.	6	.	2
6	.	.	3	.	1

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$5 \bullet_7 2 =$	3
$6 \bullet_7 2 =$	5
$3 \bullet_7 2 =$	6
$5 \bullet_7 3 =$	1
$6 \bullet_7 3 =$	4
$3 \bullet_7 3 =$	2
$5 \bullet_7 5 =$	4
$6 \bullet_7 5 =$	2
$3 \bullet_7 5 =$	1

Your last answer was interpreted as follows:

3

Your last answer was interpreted as follows:

5

Your last answer was interpreted as follows:

6

Your last answer was interpreted as follows:

1

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

2

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

2

Your last answer was interpreted as follows:

1

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 552

Signing code:

Your last answer was interpreted as follows:

38614

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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